

# Sang Min Lee

*Ph.D. Candidate in Artificial Intelligence · Structural & Wind Engineering*  
*Seoul, Republic of Korea*

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## Research Interests

- AI for wind engineering: wind load, pressure, and force estimation; sensor reduction for wind-resistant design
- Learning-based structural assessment: health monitoring, anomaly detection, automated impact-echo evaluation
- Large language models for engineering practice: retrieval-augmented generation and mixture-of-experts for design codes
- Physics-informed and data-driven modeling of structural and material behavior

## Education

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| <b>Ph.D. in Artificial Intelligence (Interdisciplinary Program)</b> , Seoul National University                                                                                     | <i>Seoul, Republic of Korea</i> |
| <ul style="list-style-type: none"><li>• Dissertation: “Deep Learning Application for Wind-Resistant Design of Building Structure”</li><li>• Advisor: Prof. Thomas Kang</li></ul>    | <i>Mar. 2023 - Feb. 2027</i>    |
| <b>M.S. in Architecture and Architectural Engineering</b> , Seoul National University                                                                                               | <i>Seoul, Republic of Korea</i> |
| <ul style="list-style-type: none"><li>• Thesis: “Machine Learning Application to Automated Impact-Echo Test for Concrete Structures”</li><li>• Advisor: Prof. Thomas Kang</li></ul> | <i>Mar. 2021 - Feb. 2023</i>    |
| <b>B.S. in Electrical and Computer Engineering (Double Major)</b> , Seoul National University                                                                                       | <i>Seoul, Republic of Korea</i> |
|                                                                                                                                                                                     | <i>Sep. 2017 - Feb. 2021</i>    |
| <b>B.S. in Architecture and Architectural Engineering</b> , Seoul National University                                                                                               | <i>Seoul, Republic of Korea</i> |
|                                                                                                                                                                                     | <i>Mar. 2016 - Feb. 2021</i>    |
| <b>Science High School for the Gifted</b> , Korea Science Academy of KAIST                                                                                                          | <i>Busan, Republic of Korea</i> |
|                                                                                                                                                                                     | <i>Mar. 2013 - Feb. 2016</i>    |

## Experience

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| <b>Visiting Student Researcher</b> , The University of Oklahoma                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Norman, OK, USA</i>          |
| <ul style="list-style-type: none"><li>• School of Civil Engineering and Environmental Sciences &amp; National Weather Center</li></ul>                                                                                                                                                                                                                                                                                                                                                 | <i>Oct. 2025 - Nov. 2025</i>    |
| <b>Conference Organizing Committee Member</b> , Wind Engineering Institute of Korea                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Seoul, Republic of Korea</i> |
| <ul style="list-style-type: none"><li>• Organizing committee, 28th WEIK Annual Conference &amp; General Assembly (Seoul National University)</li></ul>                                                                                                                                                                                                                                                                                                                                 | <i>Apr. 2025 - May 2025</i>     |
| <b>Academic Reviewer</b> , Wind and Structures (peer-reviewed SCIE journal)                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |
| <ul style="list-style-type: none"><li>• Volunteer reviewer in the field of wind engineering</li></ul>                                                                                                                                                                                                                                                                                                                                                                                  | <i>Jan. 2024 - Present</i>      |
| <b>Graduate Research Assistant</b> , Structural Engineering Laboratory, Seoul National University                                                                                                                                                                                                                                                                                                                                                                                      | <i>Seoul, Republic of Korea</i> |
| <ul style="list-style-type: none"><li>• Research projects (mainly participated)<ul style="list-style-type: none"><li>- Hyper-converged Forensic Research Center for Infrastructure (2021~2027, NRF/MSIT)</li><li>- Smart City Innovative Technology Demonstration (2024~2025, KAIA/MOLIT)</li><li>- Outstanding Patent Prototype Development (2023, Seoul National University)</li><li>- AI for Construction (2024, Seoul National University/Sejong Construction)</li></ul></li></ul> | <i>Mar. 2021 - Feb. 2027</i>    |
| <b>Graduate Teaching Assistant</b> , Seoul National University                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Seoul, Republic of Korea</i> |
| <ul style="list-style-type: none"><li>• Architecture and AI (22S, 23S, 24S)<ul style="list-style-type: none"><li>- Guest lecturer on hands-on deep neural networks</li></ul></li><li>• Introduction to IoT, AI, and Big Data (21S, 21F, 22S, 22F)<ul style="list-style-type: none"><li>- IoT project evaluator</li><li>- Arduino UNO</li><li>- Head TA (22F)</li></ul></li><li>• Creative Engineering Design (21F, 22S)</li></ul>                                                      | <i>Mar. 2021 - Jun. 2024</i>    |

- IoT project evaluator
- Arduino UNO, ESP32
- IoT and Creative Engineering Design (21F, 22S)
  - IoT project evaluator
  - Arduino UNO, Teensy, ESP32, STM boards
- Structural Experiment and Material (22S)
  - Euler buckling, truss deflection, shake-table seismic test
- Mechanics of Materials for Architectural Engineering (21F)
  - Fundamentals for structural engineering
- Prestressed Concrete (21F)
  - Class management for post-tensioned structures
- Computation in Architectural Engineering (21S)
  - MATLAB for structural engineering students

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**Undergraduate Research Assistant**, Seoul National University

*Seoul, Republic of Korea*

- HpSE, Seoul National University (2021)  
wind pressure coefficient prediction for wind-resistant building design
- SNUCEM (2019~2020)  
AHP analysis for improvement of military dining facilities
- AEPRL, Seoul National University (2018~2019)  
IoT device (hardware and software) for indoor surface-condensation detection

*Dec. 2018 - Feb. 2021*

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**Research Engineer (Intern)**, Chungyeon Architects

*Seoul, Republic of Korea*

- Data analysis for building energy performance evaluation (ECO2, AutoCAD, Python)

*Jul. 2018 - Aug. 2018*

## **Publications**

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### **International Journal Articles**

1. **Lee, S. M.**, Alinejad, H., and Kang, T. H.-K. (2026) "Wind force time history reconstruction with reduced pressure taps using deep learning," *Engineering Applications in Artificial Intelligence*. Under review.
2. Yan, S., Gupta, H., Chen, M., Zhu, S., Li, Z., Wen, Y., Zhang, M., Cao, J., Chen, X., **Lee, S. M.**, Kang, T. H.-K., Duan, Q., Sorooshian, S., and Hong, Y. (2026) "Towards a reasoning-based physics-informed framework for efficient parameter specification in earth system modeling," *Science Advances*. Under review.
3. **Lee, S. M.**, Yan, S., Hong, Y., and Kang, T. H.-K. (2026) "Synoptic and non-synoptic wind classification using pseudo-labeling and machine learning with the Oklahoma Mesonet database," *Journal of Wind Engineering and Industrial Aerodynamics*. Under review.
4. **Lee, S. M.**, Hong, J., Choi, H., and Kang, T. H.-K. (2025) "Machine learning assisted method for automated impact-echo testing of concrete structure," *Journal of Nondestructive Evaluation*, 44(4), 121.
5. Lee, Y. I., **Lee, S. M.**, and Kang, T. H.-K. (2025) "Multiclass deep support vector data description for structural health monitoring," *ASCE Journal of Structural Engineering*, 151(12).
6. **Lee, S. M.**, Choi, H.-S., Kim, C., and Kang, T. H.-K. (2025) "Lightweight alternative machine learning model for automating concrete crack image classification," *ACI Structural Journal*, 122(5).

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### **Domestic Journal Articles**

1. Ahn, B., Lee, D., **Lee, S. M.**, and Kang, T. H.-K. (2023) "Research on satellite image-based wind speed prediction methodology using neural network," *Journal of Wind Engineering Institute of Korea*, 27(4), 135–142. [In Korean]
2. **Lee, S. M.**, Sung, H. S., and Kang, T. H.-K. (2022) "Comparison of performance for predicting compressive strength of concrete using machine learning," *Journal of the Korea Concrete Institute*, 34(5), 505–513. [In Korean]

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### **International Conference Proceedings**

1. **Lee, S. M.** and Kang, T. (2025) "Retrieval augmented generation for assistance of using wind load design codes," *The 10th Asia-Pacific Conference on Wind Engineering (APCWE10)*, Chengdu, China.

2. **Lee, S. M.**, Lee, Y. I., and Kang, T. (2024) “Multi-sphere deep SVDD for system identification of building structure,” *The 2024 World Congress on Advances in Civil, Environmental, & Materials Research*, Seoul, Republic of Korea.
3. **Lee, S. M.**, Choi, H.-S., Kim, C., and Kang, T. (2024) “Representation learning for classification of concrete crack images,” *The 2024 Structures Congress*, Seoul, Republic of Korea.
4. **Lee, S. M.** and Kang, T. (2022) “Clustering wind pressure tap using dynamic time warping,” *8th European-African Conference on Wind Engineering (EACWE2022)*, Bucharest, Romania.
5. **Lee, S. M.** and Kang, T. (2022) “Shear strength prediction of prestressed concrete based on machine learning,” *The 2022 Structures Congress*, Seoul, Republic of Korea.
6. **Lee, S. M.** and Kang, T. (2021) “Prediction of wind pressure coefficients on high-rise building façade using LSTM RNN model for sensor reduction,” *The 2021 World Congress on Advances in Structural Engineering and Mechanics*, Seoul, Republic of Korea.

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#### Domestic Conference Proceedings

1. **Lee, S. M.** and Kang, T. (2026) “Deep learning-based reconstruction of wind force time history from reduced pressure tap measurements,” *The Wind Engineering Institute of Korea 2026 Conference*, Gwangju, Republic of Korea. [In Korean]
2. **Lee, S. M.** and Kang, T. (2026) “Development of edge AI impact echo system for internal defect inspection in concrete structure,” *Korea Concrete Institute 2026 Spring Conference*, Changwon, Republic of Korea. [In Korean]
3. Park, H. G., **Lee, S. M.**, and Kang, T. (2026) “Application of impact-echo testing to liner-plate-backed concrete with simulated internal defects in nuclear containment buildings,” *Korea Concrete Institute 2026 Spring Conference*, Changwon, Republic of Korea. [In Korean]
4. **Lee, S. M.** and Kang, T. (2025) “Clustering of wind pressure coefficients on concrete building facades using deep learning,” *Korea Concrete Institute 2025 Spring Conference*, Jeju, Republic of Korea. [In Korean]
5. Lee, Y. I., **Lee, S. M.**, and Kang, T. (2024) “Structural health monitoring of concrete building using multi-class deep SVDD,” *Korea Concrete Institute 2024 Fall Conference*, Sokcho, Republic of Korea. [In Korean]
6. **Lee, S. M.** and Kang, T. (2024) “Framework of large language model for structural design code using mixture of experts,” *Korea Concrete Institute 2024 Fall Conference*, Sokcho, Republic of Korea. [In Korean]
7. **Lee, S. M.**, Choi, H.-S., Kim, C., and Kang, T. (2024) “Lightweighting concrete crack image classification machine learning models for on-device AI,” *Korea Concrete Institute 2024 Spring Conference*, Gwangju, Republic of Korea. [In Korean]
8. **Lee, S. M.**, Choi, H., and Kang, T. (2023) “Automation of impact-echo test using machine learning,” *Korea Concrete Institute 2023 Spring Conference*, Busan, Republic of Korea. [In Korean]
9. **Lee, S. M.**, Sung, H. S., and Kang, T. (2022) “Prediction of concrete compressive strength using machine learning,” *Korea Concrete Institute 2022 Spring Conference*, Jeju, Republic of Korea. [In Korean]
10. **Lee, S. M.** and Kang, T. (2021) “Prediction of wind pressure coefficient for concrete building wall using LSTM RNN,” *Korea Concrete Institute 2021 Fall Conference*, Gyeongju, Republic of Korea. [In Korean]
11. **Lee, S. M.** and Kang, T. (2021) “Prediction of wind pressure coefficients for concrete building walls using LSTM network,” *Korea Concrete Institute 2021 Spring Conference*, Yeosu, Republic of Korea. [In Korean]

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## Patents

1. **Lee, S. M.** and Kang, T. H.-K. (2025) “AI-based defect detection system inside concrete members,” EP 23915029.5 (applied).
2. **Lee, S. M.** and Kang, T. H.-K. (2024) “AI-based defect detection system inside concrete members,” US 18/875,721 (applied).
3. **Lee, S. M.** and Kang, T. H.-K. (2023) “AI-based defect detection system inside concrete members,” KR 10-2480382 (registered) & PCT/KR2023/006392 (applied).
4. **Lee, S. M.** and Kang, T. H.-K. (2022) “Wind load estimation system based on artificial intelligence,” KR 10-2611457 (registered) & PCT/KR2023/021982 (applied).

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## Honors & Awards

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|-------------------------------------------------------------------------------------------------------------------------|-------------|
| 1. <b>AI Fellowship</b> , Graduate School of Artificial Intelligence, Seoul National University                         | 2025 & 2026 |
| 2. <b>AI Paper Competition Excellence Award</b> , Graduate School of Artificial Intelligence, Seoul National University | 2025        |
| 3. <b>Graduate School of Artificial Intelligence Fellowship</b> , Seoul National University                             | 2023 & 2024 |
| 4. <b>Best Teaching Assistant Award</b> , College of Engineering, Seoul National University                             | 2022        |

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| 5. <b>Conference Outstanding Paper Award</b> , Korea Concrete Institute                                                    | 2022 |
| 6. <b>Full Scholarship (SEN Engineering Group)</b> , Dept. of Architecture & Architectural Eng., Seoul National University | 2021 |

## Licensure & Certification

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| 1. <b>Big Data Analysis Engineer (National Technical Qualification)</b> , Korea Data Agency | 2025 |
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## Professional Membership

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| 1. <b>Student Member</b> , American Society of Civil Engineers (ASCE)                       | 2025–2027 |
| 2. <b>Student Member</b> , Wind Engineering Institute of Korea (WEIK)                       | 2023–2027 |
| 3. <b>Student Member</b> , Korea Institute for Structural Maintenance and Inspection (KSMI) | 2023–2027 |
| 4. <b>Student Member</b> , Architectural Institute of Korea (AIK)                           | 2023–2027 |
| 5. <b>Student Member</b> , American Concrete Institute (ACI)                                | 2021–2027 |
| 6. <b>Student Member</b> , Korea Concrete Institute (KCI)                                   | 2021–2027 |